Concept & Uses of Classification Systems Continued

Exam code: 0610 & 0970

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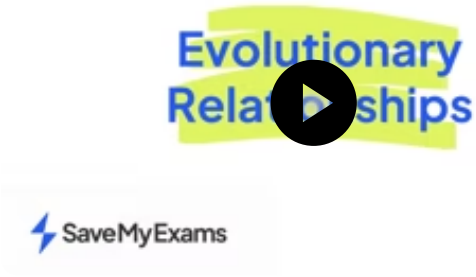
Updated on 3 October 2024



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Reflecting Evolutionary Relationships: Extended

Extended Tier Only

- Classification systems aim to **reflect evolutionary relationships** between species
- Traditional biological classification systems grouped organisms based on the **features** that they **shared**
  - If organisms shared more similar features then they were said to be more closely related
- In the past, scientists have encountered many **difficulties** when trying to determine the **evolutionary relationships** of species based on this method
- Using the physical features of species (such as colour/shape/size) has **many limitations** and can often lead to the **wrong classification** of species

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Got it

Organisms: Extended

Extended Tier Only

- Organisms share features because they originally descend from a **common ancestor**
- Example: all mammals have bodies covered in hair, feed young from mammary glands and have external ears (pinnas)
- Originally, organisms were classified using **morphology** (the overall form and shape of the organism, e.g. whether it had wings or legs) and **anatomy** (the detailed body structure as determined by dissection)
- As technology advanced, **microscopes**, knowledge of **biochemistry** and eventually **DNA sequencing** allowed us to classify organisms using a more scientific approach
- Studies of DNA sequences of different species show that the **more similar the base sequences in the DNA of two species, the more closely related those two species are** (and the more recent in time their common ancestor is)
- This means that the **base sequences in a mammal's DNA are more closely related to all other mammals** than to any other vertebrate groups

DNA sequences can show how closely related different species are

- The sequences above show that Brachinus armiger and Brachinus hirsutus are **more closely related** than any other species in the list as their DNA sequences are identical except for the last but one base (B.armiger has a T in that position whereas B.hirsutus has an A)
- As DNA base sequences are used to code for **amino acid sequences in proteins**, the similarities in amino acid sequences can also be used to determine how closely related organisms are

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